CSC662 mock paper 02 - planner revision, theory module and report

Generated locally from the 35 past papers and the read-only CSC662 lecture decks. No original paper or deck was modified.

Deliverables

- `outputs/mock_paper_02.docx` / `.pdf` - realistic-mode paper, 4 pages.
- `outputs/mock_scheme_02.docx` / `.pdf` - marking scheme, 13 pages.
- `outputs/planner_revised_realistic.json` / `.md` - revised slot table.
- `outputs/theory_drafts_realistic.json` / `.md` - verified theory questions and sources.
- `outputs/vigenere_tableau.png` - 26 x 26 tableau printed after Q2(b).
- `outputs/final_report.md` / `.pdf` - this report (7 pages in the PDF render).

1. Measurements (reported before the planner was changed)

- Clean modern papers: **10**; papers with explicit Part A/B labels used for pair/part measures: **8** (CSC662-2019-06, CSC662-2019-12, CSC662-2022-02, CSC662-2023-02, CSC662-2024-07, CSC662-2025-02, CSC662-2025-07, CSC662-2026-02).
- Coverage caveat: 2018-01 and 2018-06 use a numeric-question layout with no A/B labels and are excluded from the pair/part measures; 2025-07 has incomplete/noisy A/B extraction. 2026-02 marks are noisy in places but its valid pairs are included.

(a) Part A (a)/(b) relationship

- Pairs measured: **72**.
- Same topic: **55/72 (76.4%)**.
- (b) explicitly refers to (a) / the above / the scenario: **7/72 (9.7%)**.

Dependency regex: `your answer in (a)`, `answer in (a)`, `the above`, `based on the above`, `scenario above`, `given scenario`, `this scenario`, `same scenario`, `in part (a)`, `using your answer`.

(b) Command-word distribution by part

Label	Part A rows	Part A marks	Part B rows	Part B marks
State	31	110	7	35
List	12	28	0	0
Explain	56	198	13	54
Describe	16	84	2	8
Discuss	17	90	5	27
Compare	3	18	0	0
Justify	4	24	5	28
Other	7	30	11	44

Raw phrasing mapped to the seven labels: define -> Describe; what is -> Explain; identify/give/name -> State; why/how -> Explain; propose/suggest/support -> Justify; differentiate -> Compare. `Other` means no mapped command word and is never used as a generated slot label.

(c) Topic distribution in Part B

Topic	Rows	Marks
risk/management	9	39

Topic	Rows	Marks
attacks/malware	8	36
access control models	8	35
authentication/biometrics	5	23
software/OS security	3	15
trusted computing/blockchain	2	10
general cryptography	2	10
hashing/signatures	2	10
network security	1	6
block ciphers	1	5
public-key	1	5
other	1	2

Part B topic and task mix is used separately from Part A: the two Part B questions take the two largest measured Part B topic shares (risk/management 39 marks, attacks/malware 36), and the four Part B slots take the measured Part B 5-mark command mix (Explain 2, State 1, Discuss 1 in this seed).

2. Revised planner table (realistic mode, seed 9)

- 24 slots, 100 marks: Part A 80, Part B 20; computed marks **6** (clean-modern mean 4.0).
- Question topics: `A1:network security -> A2:classical ciphers -> A3:trusted computing/blockchain -> A4:authentication/biometrics -> A5:software/OS security -> A6:attacks/malware -> A7:access control models -> A8:attacks/malware -> A9:authentication/biometrics -> A10:access control models -> B1:risk/management (ethics) -> B2:attacks/malware`.
- One topic per question: **yes**; no more than two consecutive questions with the same topic: **PASS**.
- Dependent pair(s): **A3** (1 of 9 Part A theory pairs = 11.1%, close to the measured 9.7%).

Slot	Marks	Kind	Topic	Command	Computed type	Target size / key	Style exact / mode	Low conf.	Dependent	Notes
A1a	2	theory	network security	Explain			17 / exact			style n=17
A1b	6	theory	network security	Explain			3 / exact			style n=3
A2a	2	theory	classical ciphers	List			0 / topic+marks (any command)	yes		style n=0; widened to topic+marks (any command); low style confidence
A2b	6	computed	classical ciphers	compute	Vigenere	13-18 letters; keyword 6-8 letters				computed slot; print the 26 x 26 Vigenere tableau in the question
A3a	2	theory	trusted computing/blockchain	State			4 / exact		yes	style n=4; dependent pair
A3b	6	theory	trusted computing/blockchain	Justify			0 / topic+marks (any command)	yes	yes	style n=0; widened to topic+marks (any command); low style confidence; dependent pair
A4a	2	theory	authentication/biometrics	Explain			11 / exact			style n=11
A4b	6	theory	authentication/biometrics	Discuss			5 / exact			style n=5
A5a	2	theory	software/OS security	State			2 / topic+marks (any command)	yes		style n=2; widened to topic+marks (any command); low style confidence
A5b	6	theory	software/OS security	Describe			2 / topic+marks	yes		style n=2; widened to topic+marks (any

Slot	Marks	Kind	Topic	Command	Computed type	Target size / key	Style exact / mode	Low conf.	Dependent	Notes
							(any command)			command); low style confidence
A6a	2	theory	attacks/malware	List			0 / topic+marks (any command)	yes		style n=0; widened to topic+marks (any command); low style confidence
A6b	6	theory	attacks/malware	Describe			12 / exact			style n=12
A7a	2	theory	access control models	Explain			5 / exact			style n=5
A7b	6	theory	access control models	State			1 / topic+marks (any command)	yes		style n=1; widened to topic+marks (any command); low style confidence
A8a	2	theory	attacks/malware	State			6 / exact			style n=6
A8b	6	theory	attacks/malware	State			5 / exact			style n=5
A9a	2	theory	authentication/biometrics	Explain			11 / exact			style n=11
A9b	6	theory	authentication/biometrics	Discuss			5 / exact			style n=5
A10a	2	theory	access control models	Explain			5 / exact			style n=5
A10b	6	theory	access control models	Explain			4 / exact			style n=4
B1a	5	theory	risk/management (ethics)	Explain			2 / topic+marks (any command)	yes		style n=2; widened to topic+marks (any command); low style confidence
B1b	5	theory	risk/management (ethics)	State			2 / topic+marks (any command)	yes		style n=2; widened to topic+marks (any command); low style confidence
B2a	5	theory	attacks/malware	Explain			1 / topic+marks (any command)	yes		style n=1; widened to topic+marks (any command); low style confidence
B2b	5	theory	attacks/malware	Discuss			0 / topic+marks (any command)	yes		style n=0; widened to topic+marks (any command); low style confidence

Topic allocation vs modern share

Topic	Modern weight	Target marks	Assigned theory marks
attacks/malware	0.290	27	26
authentication/biometrics	0.189	17	16
access control models	0.147	14	16
trusted computing/blockchain	0.096	9	8
software/OS security	0.058	5	8
network security	0.068	6	8
risk/management (ethics)	0.152	14	10

The planner renormalised the seven theory topics after adding `risk/management (ethics)` (modern risk marks = 119; source deck 071 Ethics). Planned-vs-target differences are reported rather than hidden: risk is 10 vs 14 target, software/OS 8 vs 5, network 8 vs 6, access control 16 vs 14. A 2-mark theory slot paired with the computed Vigenere question is on `classical ciphers` (one-topic-per-question rule) and sits outside the seven-topic target.

Coverage note: 173 of 955 modern non-compute marks (18.1%) fall in topics outside the seven (other, general cryptography, classical ciphers, public-key, hashing/signatures, block ciphers) and were not reallocated. The ethics deck contributes 16 text chunks from 18 slides (slides 1,3-9,11-18), 6,387 characters.

3. Three real 6-mark examples per command label used at 6 marks

The labels used at 6 marks in this paper are State, Explain, Describe, Discuss and Justify. These are real past questions whose marks column is 6; noisy OCR/header text was skipped for the display.

State

- **CSC662-2009-04 Q1d**: Determine whether the P-box with the following tables is a straight P-box, a compression P-box or an expansion P-box. State one reason for each table. i) ii) iii) 112 3 44 15 3 6 78 2 14 3 56
- **CSC662-2010-10 Q11b**: Assuming that high > low, identify program fragments that are desired (allowed) or undesired (prevented) in information security flows for reading operation.
- **CSC662-2014-06 Q15a**: State THREE (3) purposes of today's implementation of steganography.

Explain

- **CSC662-2007-10 Q1a**: Explain how three goals of computer security relate with each other in building a secure system.
- **CSC662-2008-10 Q2b**: Recently, there are many types of malware infection had been reported to the security organization. Explain the characteristics of THREE (3) types of malware that you know.
- **CSC662-2009-04 Q2b**: A company is said as having a secured computer system. In your own words explain what it is meant by the term a secured computer system.

Describe

- **CSC662-2006-10 Q1a**: Describe the difference between vulnerability, threat and control, by using one example.
- **CSC662-2007-04 Q1a**: Define the following terms: i) Firewall ii) Integrity iii) Vulnerability
- **CSC662-2007-04 Q2b**: Describe the term malicious code and explain TWO (2) examples of them.

Discuss

- **CSC662-2010-04 Q11**: Discuss THREE (3) approaches that can be used to create good passwords. (6 marks)
- **CSC662-2010-10 Q3b**: Discuss THREE (3) important characteristics of 'economically feasible" security.
- **CSC662-2012-06 Q15**: Software security is a system-wide issue that involves both building in security mechanisms and designing the system to be robust. Discuss the following software security mechanism: a) Data execution protection b) Address space layout randomization c) Canary

Justify

- **CSC662-2016-12 Q11ii**: Suggest ONE (1) unique type of biometric methods for each of the situation below and give a reason. ii. iii. Entry and exit points in a nuclear research center.
- **CSC662-2025-07 QA1b**: Propose how a biometric authentication system can be integrated into the hospital's access control process. Discuss THREE (3) factors that must be considered to ensure its effecliveness and security.
- **CSC662-2025-07 QB4b**: Propose THREE (3) different methods the developer could use to prevent such code injection attacks.

4. Theory module

Pipeline

- Deck ingestion: python-pptx, read-only from the KB folder `C:/Users/Ahmad/Desktop/CSC662`; 25 decks, 496 slide chunks; every chunk keeps deck + slide number.

- Retrieval: BM25 (k1=1.5, b=0.75), top 10 chunks, max 4 per deck; topic-specific deck allow-lists (ethics restricted to deck 071). Fallback to all decks is recorded if fewer than 3 hits.
- Generation: Qwen cheap model via the existing OpenAI-compatible client with `enable_thinking=False`; Pydantic JSON schema; one call per question pair, with up to 3 retries after the initial attempt.
- Verification: quote must be a verbatim contiguous substring of the cited retrieved chunk (whitespace-normalised); ellipses rejected; exactly one marking point per mark; command word present; dependent (b) must refer back to (a); non-dependent siblings must use different angles; cross-generated duplicates rejected; fuzzy near-copy against all 35 papers rejected at SequenceMatcher >= 0.80.
- Every accepted theory answer is labelled `sourced`; each point carries deck + slide + chunk id.

Results and cost

- Model: **qwen3.7-flash**; attempt limit: 4 (initial + up to 3 retries).
- Slots filled: **24/24** (23 theory sourced, 1 computed); accepted theory questions: **23** / 23 theory slots; placeholders left: **0**.
- Rejected slots in the final artifact: **0**; intermediate rejected attempts are listed below with reasons.
- Theory marks covered: **94**; marking points: **94**; every point has a verified quote.
- Calls: **60**; tokens: 122486 prompt + 36446 completion = 158932.
- Estimated cost: **$0.0106**; cost per accepted question: **$0.0005**.
- Cost basis: public qwen-flash list prices (input $0.022/M, output $0.216/M) used as a configurable estimate; this is not an invoice.

Rejections and retries

The final pass made 28 group attempts for 12 question groups; 2 groups were accepted on the first attempt. Rejection categories in the retained attempt logs:

- point count != marks: 12
- quote too short: 9
- question length: 2
- quote not verbatim: 1

Rejected attempts (examples from the retained logs)

Question	Slots	Attempt	Rejected because
A1	A1a, A1b	1	A1b: 4 marking points for 6 marks (need exactly 6); A1b point 2: quote too short to verify
A1	A1a, A1b	2	A1a: 4 marking points for 2 marks (need exactly 2); A1a point 2: quote too short to verify
A1	A1a, A1b	3	A1b: 4 marking points for 6 marks (need exactly 6)
A10	A10a, A10b	1	A10a point 2: quote is not verbatim in 023 - Access Control and Authorization slide 4
A3	A3a, A3b	1	A3b point 6: quote too short to verify
A3	A3a, A3b	2	A3b: 3 marking points for 6 marks (need exactly 6)
A4	A4a, A4b	1	A4b point 1: quote too short to verify
A4	A4a, A4b	2	A4b point 4: quote too short to verify; A4b point 5: quote too short to verify
A4	A4a, A4b	3	A4b point 5: quote too short to verify; A4b point 6: quote too short to verify
A5	A5a, A5b	1	A5b: 3 marking points for 6 marks (need exactly 6)
A6	A6a, A6b	1	A6b: 3 marking points for 6 marks (need exactly 6)
A6	A6a, A6b	2	A6a: 4 marking points for 2 marks (need exactly 2)
A7	A7a, A7b	1	A7b: 3 marking points for 6 marks (need exactly 6)
A8	A8a, A8b	1	A8a: question too short (5 words); A8b: question too short (7 words)
B1	B1a, B1b	1	B1a: 3 marking points for 5 marks (need exactly 5); B1b: 3 marking points for 5 marks (need exactly 5)
B2	B2a, B2b	1	B2a: 2 marking points for 5 marks (need exactly 5); B2b: 2 marking points for 5 marks (need exactly 5)

The first exploratory pass and the placeholder retries also rejected near-copies, ellipsis quotes, 10-character schema failures and duplicate sibling angles; those were fixed by the chunk-id cleanup, quote-length rule, sibling/duplicate checks and per-slot angle hints. No placeholder remains in the final paper.

Attempts per question group

Question	Slots	Attempts	Result	Top retrieval
A1	A1a, A1b	4	accepted	061 - Network Security s4; 062 - Intrusion Detection System s8
A10	A10a, A10b	2	accepted	024 - Formal Security Model s5; 023 - Access Control and Authorization s18
A2	A2a	1	accepted	031 - Classical Cryptography s25; 031 - Classical Cryptography s26
A3	A3a, A3b	3	accepted	073 - Trusted Computing s8; 073 - Trusted Computing s6
A4	A4a, A4b	4	accepted	022 - Identification and Authentication s16; 022 - Identification and Authentication s8
A5	A5a, A5b	2	accepted	051 - Software Security s22; 051 - Software Security s23
A6	A6a, A6b	3	accepted	011 - Introduction to Computer Security s10; 022 - Identification and Authentication s11
A7	A7a, A7b	2	accepted	024 - Formal Security Model s5; 023 - Access Control and Authorization s18
A8	A8a, A8b	2	accepted	011 - Introduction to Computer Security s10; 022 - Identification and Authentication s11
A9	A9a, A9b	1	accepted	022 - Identification and Authentication s16; 022 - Identification and Authentication s19
B1	B1a, B1b	2	accepted	071 - Ethics and Computer Security s5; 071 - Ethics and Computer Security s6
B2	B2a, B2b	2	accepted	011 - Introduction to Computer Security s10; 022 - Identification and Authentication s11

Low style-confidence slots

Exact style examples n<3: **10 slots** - A2a, A3b, A5a, A5b, A6a, A7b, B1a, B1b, B2a, B2b. These were widened in the order exact -> same topic/any command -> same command/any topic; every accepted question still has at least 3 style examples after widening and is marked `low_style_confidence` in the JSON and scheme.

Source coverage

- 022 - Identification and Authentication: 16 marking points
- 051 - Software Security: 15 marking points
- 023 - Access Control and Authorization: 14 marking points
- 071 - Ethics and Computer Security: 10 marking points
- 073 - Trusted Computing: 8 marking points
- 062 - Intrusion Detection System: 6 marking points
- 041 - Windows Security: 6 marking points
- 072 - Biometrics and Authentication: 5 marking points
- 052 - Web Security: 5 marking points
- 061 - Network Security: 2 marking points
- 031 - Classical Cryptography: 2 marking points
- 011 - Introduction to Computer Security: 2 marking points
- 024 - Formal Security Model: 2 marking points
- 042 - Linux Security: 1 marking points

Near-copy maximum against the 35 past papers: **0.69** (threshold 0.80).

5. Paper and scheme QA

- Paper PDF: **4 pages**; scheme PDF: **13 pages**.
- Paper marks audit: 24 mark lines summing to 100 (Part A 80 + Part B 20).
- Paper headings: PART A (80 MARKS) and PART B (20 MARKS).
- Vigenere computed question: ciphertext `UMRBVVNWVPREQE`, key `CIPHER`, answer `SECURE LOGIN NOW`; the 26 x 26 tableau is printed immediately after Q2(b), titled `Vigenere tableau (A = 0)`.
- Scheme: 23 sourced theory answers + 1 computed answer + 0 placeholders; each theory point shows its quote, deck, slide and chunk id; low-style slots are flagged.
- Template checks: crest media `word/media/image1.jpeg` retained; header `CS/MOCK OCT 2026/CSC662` and footer `CONFIDENTIAL` / `Hak Cipta Universiti Teknologi MARA` retained in both paper and scheme.
- Tests: 112 pytest tests pass (`uv run pytest -q`), including planner invariants, theory quote verification, near-copy detection and dependent/sibling checks.

6. Caveats / still unverified

- The 2018 papers have no A/B labels, and 2025-07/2026-02 contain noisy extraction; measured rates use the 8 papers with explicit A/B labels (72 Part A pairs).
- Low style confidence: 10 slots lack 3 exact (topic+command+marks) examples; widening is recorded per slot.
- Some marking points quote short slide headings (e.g. `Operating System Independence`) rather than full sentences; the quote is still verbatim, but a human marker may want a fuller sentence.
- A6(b) is a password-threat question under the attacks/malware topic; it is an attack-on-authentication angle, sourced from deck 022 rather than the malware slides. The paper does not print topic labels; the scheme shows the sources.
- Playfair and affine remain older-style only and are not used in realistic mode; OTP remains unverified against the inconsistent slide 17 and is not used here.
- Cost is an estimate from public qwen-flash prices, not an account invoice; token counts are from the provider usage fields.

7. Generated files

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